

VR Project Design Document

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1 App Info

Tentative Title:	Color Room VR		
	Education & Training		Mental Health & Fitness
	Travel & Discovery		Media & Entertainment
	Productivity & Collaboration		Gaming
✓	Art & Creativity		Other: _____

2 Pitch

The goal is for users to experience and play:
in a calming and interactive virtual environment where they bring a monochromatic room to life through color-based interactions.

This will be especially memorable and fun in VR b/c:
users feel a direct sense of transformation—turning static, grey objects into vivid, animated elements using simple, intuitive controls. The immersive feedback (lights, sounds, movement) enhances emotional connection.

At a high level, during the app, users will:
Select colors from a palette, point at different parts of a stylized room, and apply color using a single button press. As they paint, objects respond dynamically—lamps turn on, music plays, curtains move, and hidden elements are revealed—creating a sense of discovery and delight.

This experience will be targeted at devices with:

6	degrees of freedom, giving users control over the	movement & rotation	of their head & controllers.
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3 Basics

The app will take place in:	and the user will get around the scene with:
a stylized, low-poly room (like a scale model) with static furniture and decorative elements.	N/A - user will be stationary movement.

The user will be able to grab:	There will not be sockets:
- The color palette (floating UI, attached to the left hand or pinned in the world)	-

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Events & Interactions

There will be haptic / audio feedback when:

- A color is successfully applied to a surface
- A reactive object is activated (e.g., candles light up, the fan starts spinning, the cat meows)

There will also be 3D sound from:

- Ambient music
- Music or sounds triggered by painting certain objects (e.g., radio, candles, cat)
- UI and room feedback sounds (button clicks, light switch, room-complete fanfare)

If the user is holding:

the color selection ray	and presses the trigger,	The targeted object or section is painted and may trigger an animation/sound/visual change
the color selection ray	and presses the trigger,	Hidden elements are revealed (e.g., framed pictures fade in) and, once the whole room is painted, the room-complete panel with confetti appears
		Suggestions: a UI change, a sound/video plays, a particle plays, an object is spawned or destroyed.

By default, the left hand will have a:

N/A	interactor.
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and the right hand will have a:

Ray	interactor.
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And you will not be able to toggle on a [Direct | Ray] interactor using the [thumbstick | button].

The main menu will be located:

as a floating palette panel attached to the left hand, or pinned in front of the user (switched with the left primary button or a toggle on the palette).

and from the main menu, the user will be able to:

- Replay the tutorial
- Restart the room (from the room-complete panel)

[Optional] There will be additional UI elements for:

- Color selection (palette) and paint progress counter

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Optimization & Publishing

To make the user experience more accessible / comfortable:

- Keep the user stationary to avoid motion sickness.
- The whole app can be played one handed: aiming, painting and UI use the right controller only
- Provide immediate visual/audio feedback for interactions.

Given that this app targets the Meta Quest 3 (not yet validated on a headset), target metrics are:

Frames per second:	≥ 72	FPS
Milliseconds per frame:	< 13.8	ms (= 1,000 / 72)
Triangles per frame:	300K - 500K	tris
Draw calls per frame:	150 - 175	batches

Lighting strategy:

☐	All baked	☒	Mostly baked with some mixed	☐	All real-time
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Light probes will also be used for more realistic mixed lighting.

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Other
features
(Optional)

- **Reactive World System:** Painting certain elements not only changes color but also triggers animations, audio, lighting changes, or environmental effects—adding layers of discovery and satisfaction. Implemented reactions: candles light up and flicker, lamps and light strings turn on, the fan spins, the radio floats music notes, the clock hands sync with real time, the cat meows, fish swim and framed pictures fade in.
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7
Sketch
(Optional)

8
Timeline
(Optional)

	Milestone	Date
1	Room models, prefabs and scene set-up	
2	Core painting, persistence and outline highlight	
3	Color palette UI and VR controller rig	
4	Unity 6 upgrade, palette pinning, progress counter and tutorial	
5	Paint reactions, juice, audio, testing, README and credits	